

A topologically principal partial answer to the twisted RD_p question

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Source question

Austad–Ortega–Palmstrom, *Polynomial growth and property RD_p for etale groupoids with applications to K -theory*, arXiv:2304.14458, define σ -twisted property RD_p for a continuous normalized 2-cocycle σ on an etale groupoid $\mathcal{G}$. They note that ordinary RD_p implies σ -twisted RD_p , and ask whether the converse holds when σ is not cohomologous to the trivial cocycle.

Result

Theorem 1. *Let $l : \mathcal{G} \rightarrow [0, \infty)$ be a length function, let $\sigma : \mathcal{G}^{(2)} \rightarrow \mathbb{T}$ be a continuous normalized 2-cocycle, and let $1 \leq p < \infty$. If either*

- 1. $\mathcal{G}$ is principal etale and l is continuous, or*
- 2. $\mathcal{G}$ is σ -compact, topologically principal, etale, and l is continuous and proper,*

then σ -twisted property RD_p with respect to l implies ordinary property RD_p with respect to l .

More precisely, for $1 < p < \infty$, σ -twisted RD_p forces polynomial growth of $\mathcal{G}$ with respect to l in either case.

Proof intuition

Weygandt proved the corresponding $p = 2$ obstruction for twisted C^* -algebras of principal and topologically principal etale groupoids. The operator-norm obstruction is not Hilbert-specific. If a source-fiber ball contains a large finite set F on which the range map is injective, the products $\gamma\eta^{-1}$, $\gamma, \eta \in F$, are uniquely labelled by the pair (γ, η) . One can therefore choose the values of a compactly supported twisted test function so that all cocycle phases cancel on the $F \times F$ block.

On the normalized vector $|F|^{-1/p} \chi_F \in \ell^p(\mathcal{G}_u)$, this test function has operator norm at least $|F|$. But its weighted ℓ^p -norm is only of order $|F|^{1/p}$ times a polynomial in the radius. Twisted RD_p therefore forbids super-polynomial growth whenever $p > 1$.

Proof

For $p = 1$, ordinary RD_1 is automatic for every etale groupoid, since the regular representation norm is bounded by the I -norm, i.e. by $\|\cdot\|_{1,0}$. Assume henceforth $1 < p < \infty$, and let q be the conjugate exponent.

Suppose, toward a contradiction, that $\mathcal{G}$ has σ -twisted RD_p with constants $A > 0$ and $k \in \mathbb{N}_0$, but does not have polynomial growth with respect to l . Then, for arbitrarily large choices of constants, there are a unit u , a radius $R \geq 1$, and a finite set

$$F \subset \{\gamma \in \mathcal{G}_u : l(\gamma) \leq R\}$$

such that

$$|F|^{1/q} > A(1 + 3R)^k. \quad (1)$$

If the ball is infinite, choose a finite subset with this cardinality. Since $\mathcal{G}$ is principal, the range map is injective on each source fiber: if $\gamma, \eta \in \mathcal{G}_u$ and $r(\gamma) = r(\eta)$, then $\gamma^{-1}\eta$ is isotropy at u , hence $\gamma = \eta$.

Put $Z = FF^{-1}$. For each $g \in Z$ there is a unique pair $(\gamma_g, \eta_g) \in F \times F$ such that $g = \gamma_g \eta_g^{-1}$. As in Weygandt's finite-ball lemma, choose pairwise small precompact open bisection neighborhoods U_g of g , with the following two properties:

$$l(h) \leq 3R \quad (h \in U_g),$$

and every source fiber and every range fiber meets $\bigcup_{g \in Z} U_g$ in at most $|F|$ points. The latter follows by first choosing bisection neighborhoods of the elements of F with pairwise disjoint range sets; then a point in a source fiber can lie over at most one fixed second coordinate $\eta \in F$, and at most $|F|$ choices of the first coordinate $\gamma \in F$.

Choose functions $\phi_g \in C_c(U_g)$ with $|\phi_g| \leq 1$ and $\phi_g(g) = 1$. Define

$$f = \sum_{g \in Z} \overline{\sigma(g, \eta_g)} \phi_g \in C_c(\mathcal{G}, \sigma).$$

For every unit v , the preceding fiber-counting estimate gives

$$\left(\sum_{h \in \mathcal{G}_v} |f(h)|^p (1 + l(h))^{pk} \right)^{1/p} \leq |F|^{1/p} (1 + 3R)^k,$$

and the same estimate on range fibers. Hence

$$\|f\|_{p,k} \leq |F|^{1/p} (1 + 3R)^k. \quad (2)$$

Let $\xi = |F|^{-1/p} \chi_F \in \ell^p(\mathcal{G}_u)$. Then $\|\xi\|_p = 1$. For $\gamma \in F$,

$$\begin{aligned} (\lambda_u^\sigma(f)\xi)(\gamma) &= |F|^{-1/p} \sum_{\eta \in F} f(\gamma\eta^{-1}) \sigma(\gamma\eta^{-1}, \eta) \\ &= |F|^{-1/p} \sum_{\eta \in F} 1 = |F|^{1-1/p}. \end{aligned}$$

Therefore

$$\|\lambda_u^\sigma(f)\xi\|_p \geq (|F| \cdot |F|^{p-1})^{1/p} = |F|,$$

so $\|f\|_{F_\lambda^p(\mathcal{G}, \sigma)} \geq |F|$. Twisted RD_p , (2), and (1) now give the contradiction

$$|F| \leq A\|f\|_{p,k} \leq A|F|^{1/p} (1 + 3R)^k < |F|.$$

Thus a principal $\mathcal{G}$ has polynomial growth with respect to l .

For the topologically principal, σ -compact, proper-length case, use Weygandt's local-slice argument in place of principality. If polynomial growth fails, properness and σ -compactness allow one to choose a large ball whose cardinality is locally constant near a unit with trivial isotropy. Taking that unit as u , the range map is injective on the finite ball in $\mathcal{G}_u$, so the same $F \times F$ construction above contradicts twisted RD_p . This is exactly Weygandt's proof of Theorem 3.3 with the ℓ^2 -normalized vector replaced by $|F|^{-1/p}\chi_F$.

Austad–Ortega–Palmstrom prove that polynomial growth implies ordinary RD_p for every $p \in [1, \infty)$. Hence $\mathcal{G}$ has ordinary RD_p , as claimed. $\square$

Verification and limitations

The proof is a direct L^p -adaptation of Weygandt's finite-ball obstruction. The only new bookkeeping is the normalization $|F|^{-1/p}\chi_F$, which keeps the lower operator bound equal to $|F|$ and changes the upper weighted norm to $|F|^{1/p}$ times a polynomial factor.

This is only a partial answer to the source question. It does not settle arbitrary étale groupoids. In fact, the arbitrary one-unit groupoid case is already Chatterji's 2006 open question: can a finitely generated group fail ordinary RD while having σ -twisted RD for some multiplier σ ? No answer to that group question was found in the bounded literature search, so this packet deliberately stops at the topologically-principal families where Weygandt's injective-range finite-ball obstruction applies.

References

- A. Austad, E. Ortega, M. Palmstrom, *Polynomial growth and property RD_p for étale groupoids with applications to K -theory*, arXiv:2304.14458.
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- V. Mathai, *Heat kernels and the range of the trace on completions of twisted group algebras*, with an appendix by I. Chatterji, arXiv:math/0606790.